7. Although the Group 0 gases are unreactive elements, some Group 0 compounds have been made.

The heavier Group 0 gases have more electron shells than the lighter ones. Their outermost electrons are shielded from the attraction of the nucleus by the inner electrons. These atoms can therefore form bonds with very reactive atoms under certain conditions.

As a rule, Group 0 gases having an atomic radius greater than 80 picometres (pm) can form compounds. The atomic radii of Group 0 elements are shown in **Table 1**.

The most common compounds of Group 0 gases are those formed with Group 7 elements. For example, krypton difluoride (KrF_2) and xenon dichloride ($XeCl_2$).

The stability of these compounds increases with increasing atomic number of the Group 0 gas and decreasing atomic number of the Group 7 element.

The oxidation state of the Group 0 gas in these compounds varies. Oxidation state is linked to the number of bonds formed. The melting points of these compounds decrease as the oxidation state of the Group 0 gas increases. An example of this is shown in **Table 2**.

Table 1

Group 0 gas	Atomic radius (pm)
He	31
Ne	38
Ar	71
Kr	88
Xe	108
Rn	120

Table 2

Xenon compound	Oxidation state of xenon	Melting point (°C)
XeF_2	+2	129
XeF_4	+4	117
XeF_6	+6	49

(a) (i) Tick (✓) the box next to the three Group 0 gases that are most likely to form compounds. [1]

helium, argon, xenon ☐

helium, neon, argon ☐

argon, krypton, xenon ☐

krypton, xenon, radon ☐

Examiner only

(ii) Tick (✓) the box next to the most stable Group 0 compound. [1]

XeF_2 ☐

KrF_2 ☐

$XeCl_2$ ☐

$KrCl_2$ ☐

(iii) Use the trend shown in **Table 2** and the information below to suggest the most likely melting point for $KrCl_6$.

Krypton compound	Oxidation state of krypton	Melting point (°C)
$KrCl_2$	+2	98
$KrCl_4$	+4	80
$KrCl_6$	+6	

Tick (✓) the box next to your answer. [1]

23 °C ☐ 71 °C ☐ 110 °C ☐

(iv) Explain why neon does not form compounds. [2]

....

....

....

....

(b) Give **one** use of helium and explain this use in terms of its properties. [2]

....

....

7

